# Shower Shape Comparison: Nominal vs Double-Interaction Mixed MC

0 mrad (22.4% DI) and 1.5 mrad (7.9% DI) Crossing Angles

PPG12 Analysis Team (automated report)

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## 1 Introduction

In $pp$ collisions at RHIC, two independent collisions can occur in the same bunch crossing (double interaction / pileup), distorting reconstructed cluster kinematics and shower shapes. To quantify this effect, full GEANT4 double-interaction MC samples (`photon10_double`, `jet12_double`) are blended with nominal single-interaction MC at the physics-dictated fractions:

- **0 mrad** crossing angle (runs 47289–51274, $\mathcal{L} = 32.7$ pb$^{-1}$): 22.4% double interaction (cluster-weighted)
- **1.5 mrad** crossing angle (runs 51274–54000, $\mathcal{L} = 16.9$ pb$^{-1}$): 7.9% double interaction (cluster-weighted)

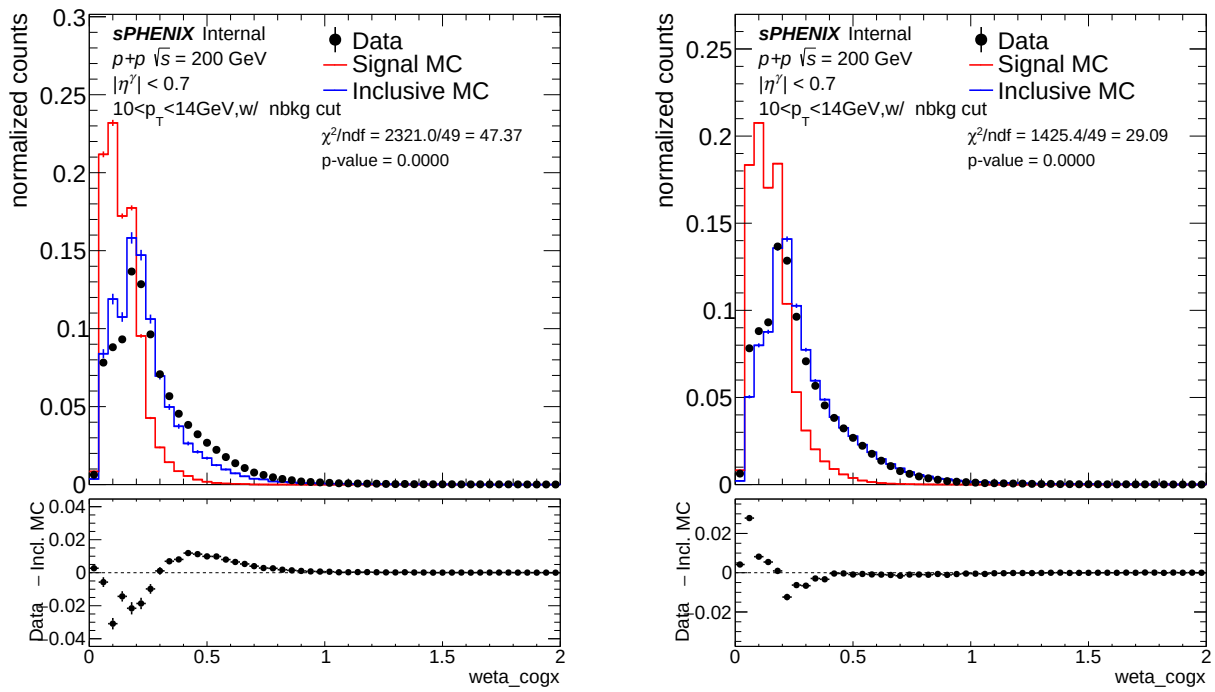
The blending uses a two-pass pipeline (`run_showershape_double.sh`): Pass 1 builds a combined vertex distribution (weighted `hadd` of single and double vtxscan files); Pass 2 reruns the full shower shape analysis with consistent vertex reweighting and `mix_weight` applied to all histogram fills. The final "mixed MC" is the `hadd` of single-weighted and double-weighted output files.

All comparisons use the **cut1** selection level (with NPB preselection). Each plot shows data (black points), signal MC (red), and inclusive MC (blue), area-normalized, with a $\chi^2$/ndf and $p$-value annotation and a Data − Inclusive MC difference panel. Left panels show nominal (single-only) MC; right panels show mixed (double-blended) MC.

## 2 0 mrad Crossing Angle (22.4% Double Interaction)

### 2.1 $w_\eta^{\mathrm{CoG}}$ (Eta Cluster Width)

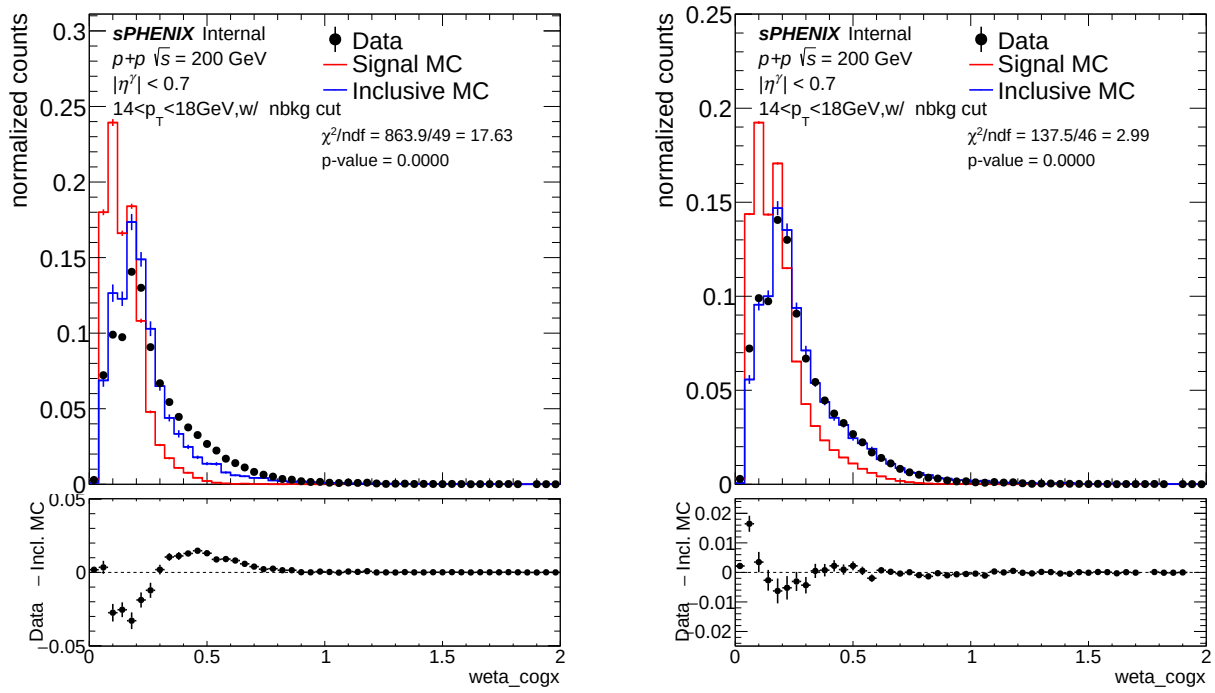
$w_\eta^{\mathrm{CoG}}$ is strongly affected by double interactions: vertex shifts displace clusters along the beam axis, broadening the $\eta$ profile.




(a) Nominal MC

(b) Mixed MC (22.4% DI)

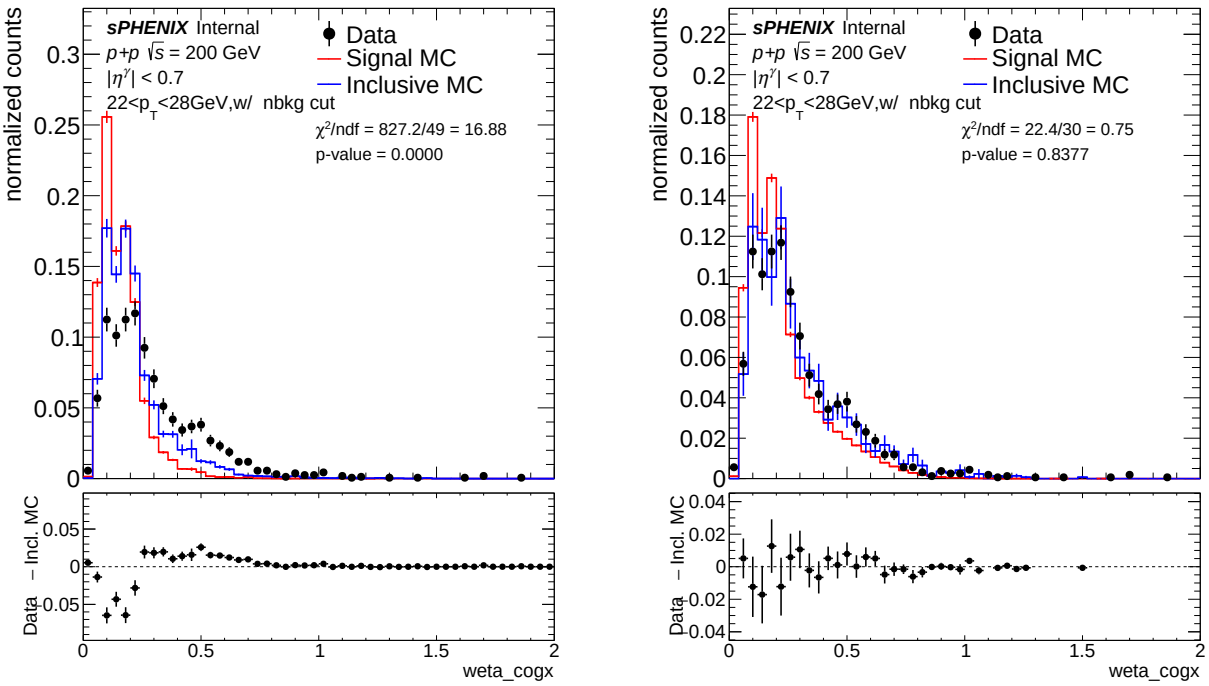
Figure 1: $w_\eta^{\mathrm{CoG}}$, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), $0\,\mathrm{mrad}$. Nominal $\chi^2/\mathrm{ndf} = 47.40$; mixed $\chi^2/\mathrm{ndf} = 28.96$ ($1.6\times$ improvement).




(a) Nominal MC

(b) Mixed MC (22.4% DI)

Figure 2: $w_\eta^{\mathrm{CoG}}$, $14 < E_T < 18\,\mathrm{GeV}$ (pt1), $0\,\mathrm{mrad}$. Nominal $\chi^2/\mathrm{ndf} = 17.64$; mixed $\chi^2/\mathrm{ndf} = 2.99$ ($5.9\times$ improvement).




(a) Nominal MC

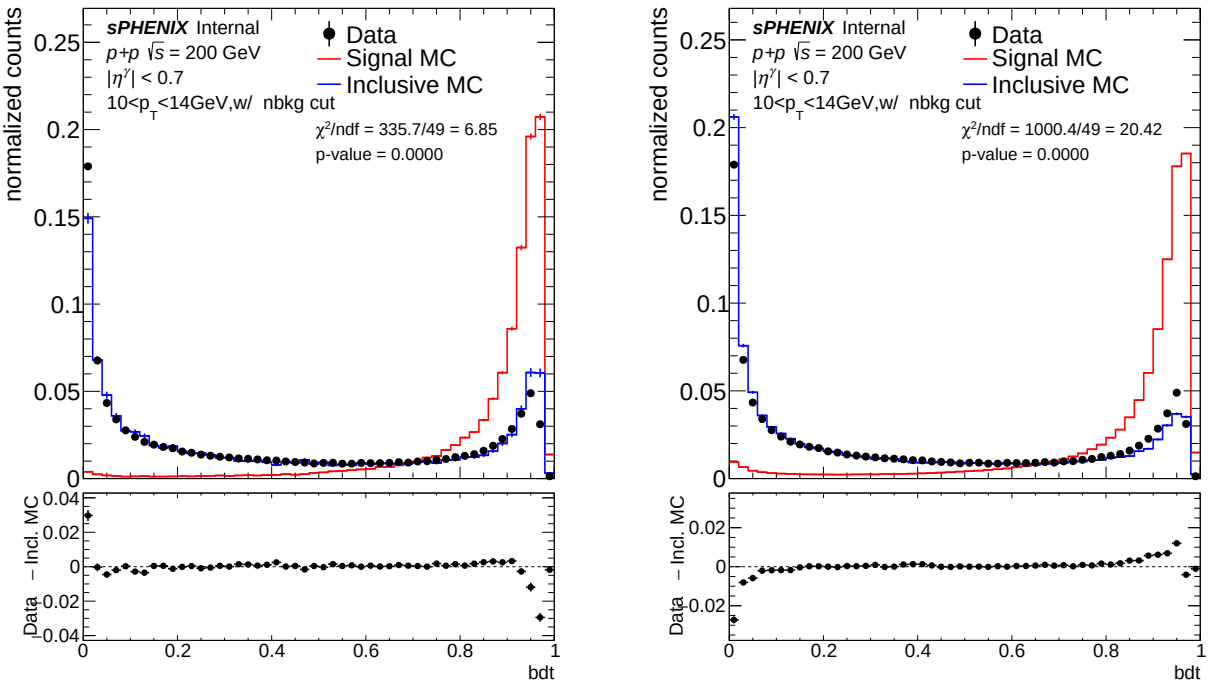
(b) Mixed MC (22.4% DI)

Figure 3: $w_\eta^{\mathrm{CoG}}$, $22 < E_T < 28\,\mathrm{GeV}$ (pt3), 0 mrad. Nominal $\chi^2/\mathrm{ndf} = 15.58$; mixed $\chi^2/\mathrm{ndf} = 0.75$ ($p = 0.84$; 20.8× improvement).

The mixed MC dramatically improves data–MC agreement for $w_\eta^{\mathrm{CoG}}$ across all $E_T$ bins at 0 mrad, with $\chi^2/\mathrm{ndf}$ dropping by factors of 1.6 (pt0) to 20.8 (pt3). At pt3, the mixed MC achieves $p = 0.84$, indicating excellent agreement.

## 2.2 BDT Score

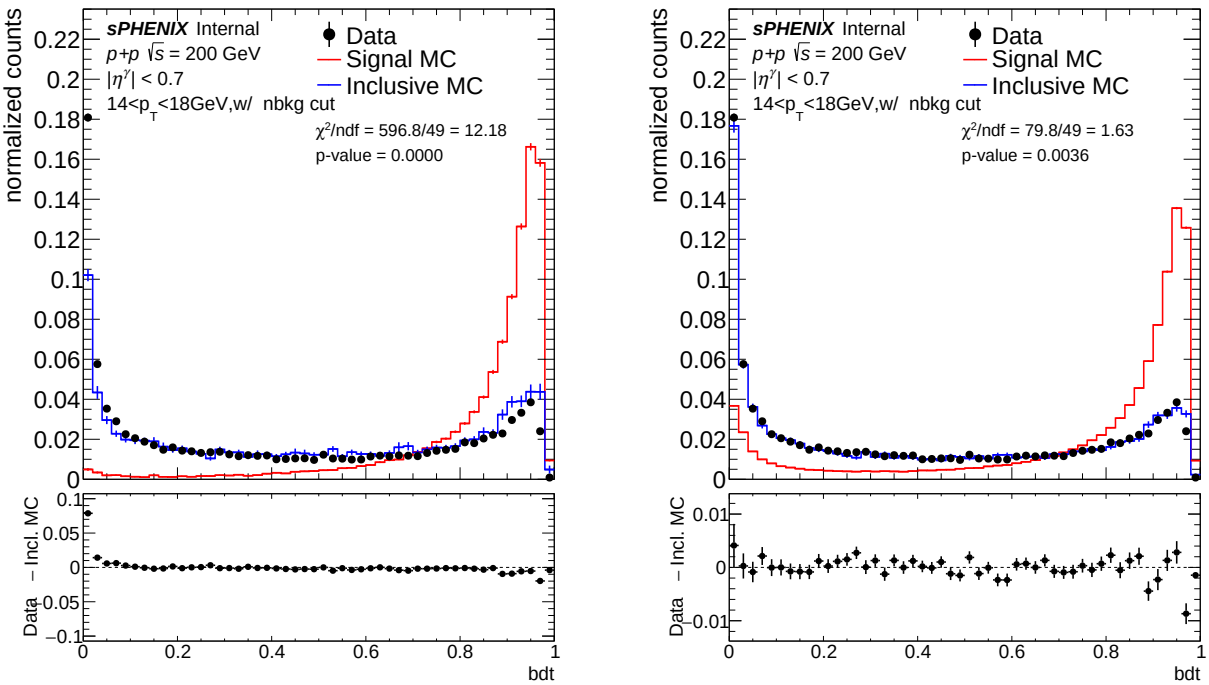
The photon/jet BDT discriminator amplifies underlying shower shape distortions through the XGBoost decision function.




(a) Nominal MC

(b) Mixed MC (22.4% DI)

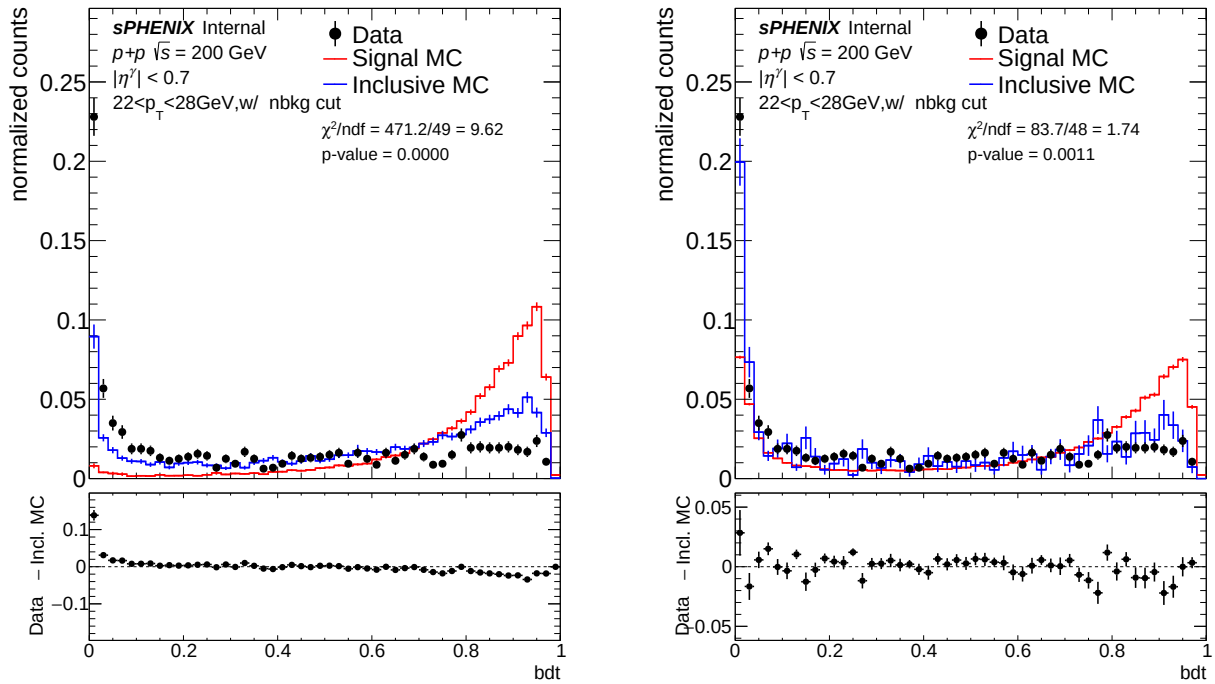
Figure 4: BDT score, $10 < E_T < 14\,\text{GeV}$ (pt0), 0 mrad. Nominal $\chi^2/\text{ndf} = 6.85$; mixed $\chi^2/\text{ndf} =$ 20.30 (3.0× **degradation**).




(a) Nominal MC

(b) Mixed MC (22.4% DI)

Figure 5: BDT score, $14 < E_T < 18\,\text{GeV}$ (pt1), 0 mrad. Nominal $\chi^2/\text{ndf} = 12.20$; mixed $\chi^2/\text{ndf} =$ 1.63 ($p = 0.004$; 7.5× improvement).



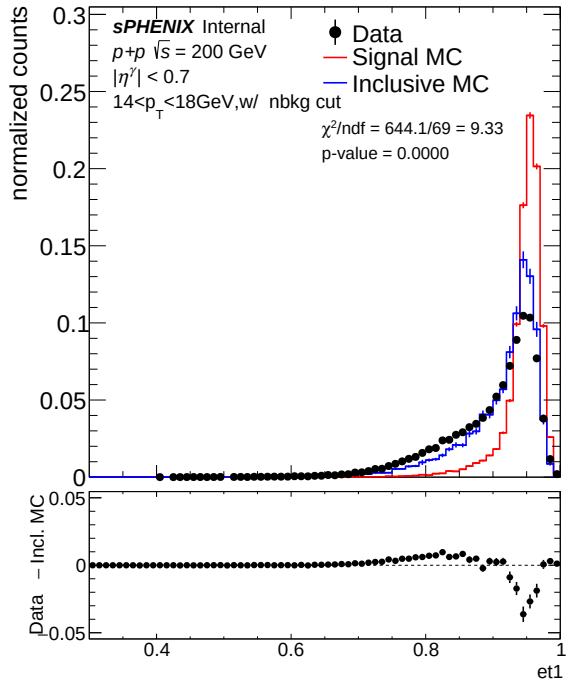

(a) Nominal MC

(b) Mixed MC (22.4% DI)

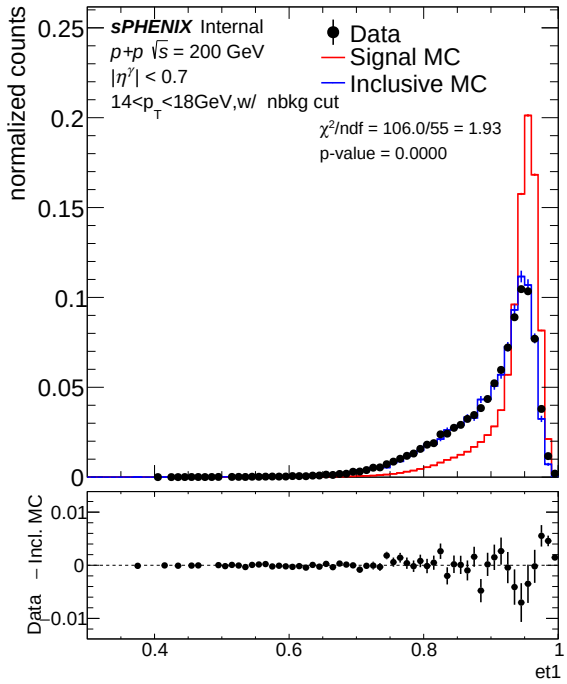
Figure 6: BDT score, $22 < E_T < 28\,\text{GeV}$ (pt3), 0 mrad. Nominal $\chi^2/\text{ndf} = 9.62$; mixed $\chi^2/\text{ndf} = 1.75$ ($p = 0.001$; 5.5× improvement).

At 0 mrad, the BDT $\chi^2/\text{ndf}$ improves substantially at pt1 (7.5×) and pt3 (5.5×). However, at pt0, the mixed MC *degrades* agreement by 3.0× (6.85 → 20.30). This low-$E_T$ degradation was not reported in the original 0 mrad analysis and is now understood to be a common feature of both crossing angles (see Section 6).

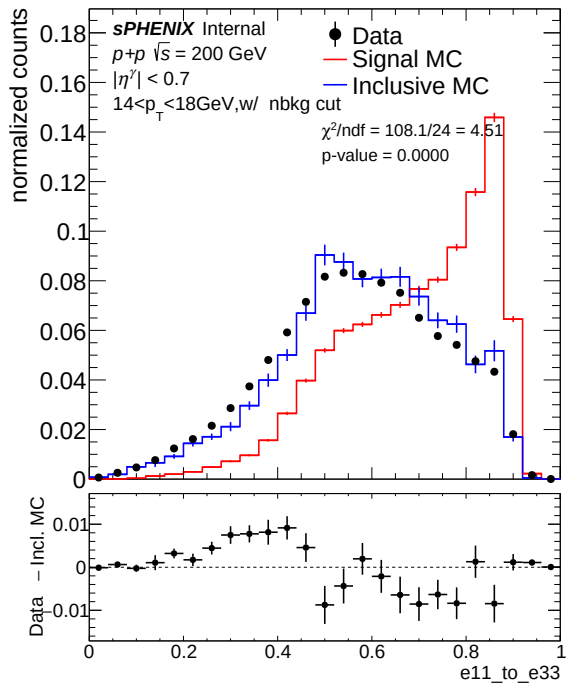
### 2.3 $f_1$ and $E_{1\times1}/E_{3\times3}$



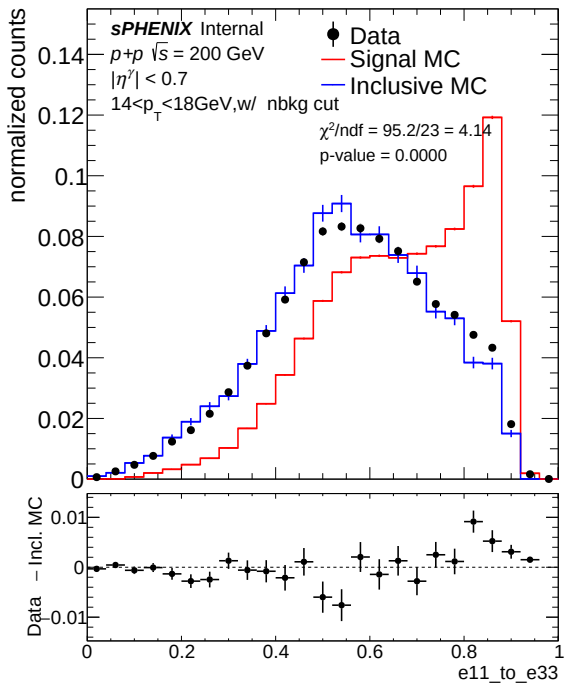

(a) $f_1$, Nominal MC




(b) $f_1$, Mixed MC (22.4% DI)




(c) $E_{1\times1}/E_{3\times3}$, Nominal MC

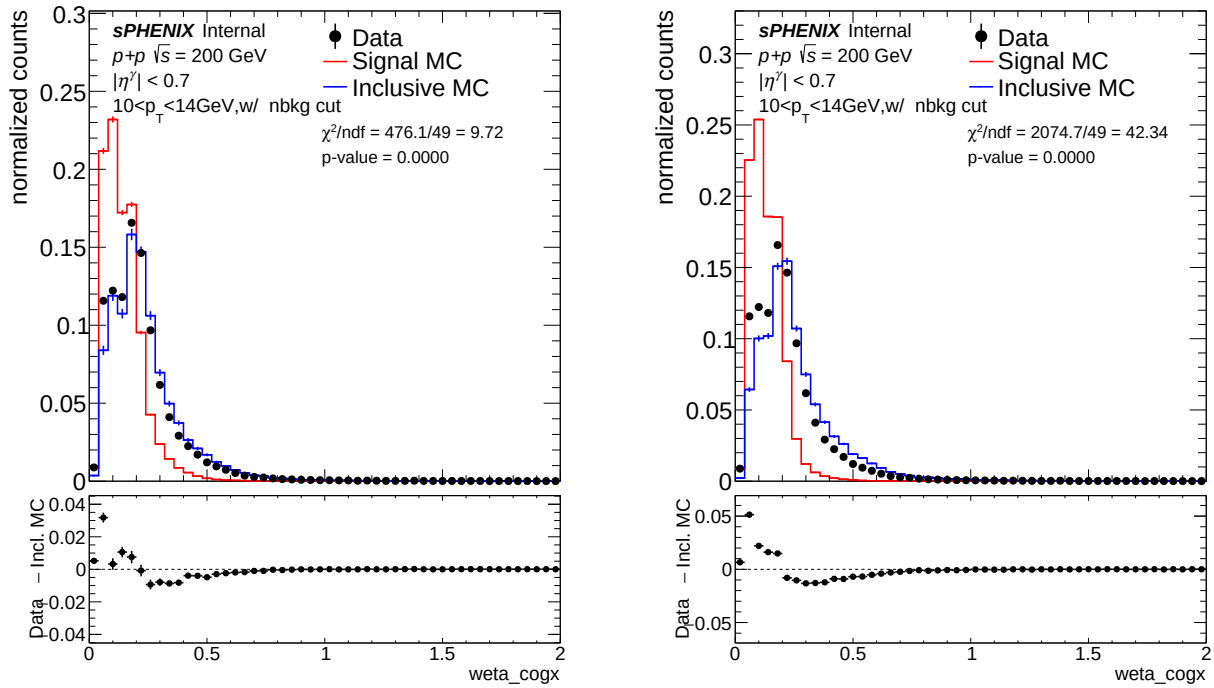



(d) $E_{1\times1}/E_{3\times3}$, Mixed MC (22.4% DI)

Figure 7: $f_1$ (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_T < 18\,\text{GeV}$ (pt1), 0 mrad. $f_1$: $\chi^2/\text{ndf} = 9.33 \rightarrow 1.92$ (4.9×). $E_{1\times1}/E_{3\times3}$: $\chi^2/\text{ndf} = 4.50 \rightarrow 4.14$ (1.1×).

## 3 1.5 mrad Crossing Angle (7.9% Double Interaction)

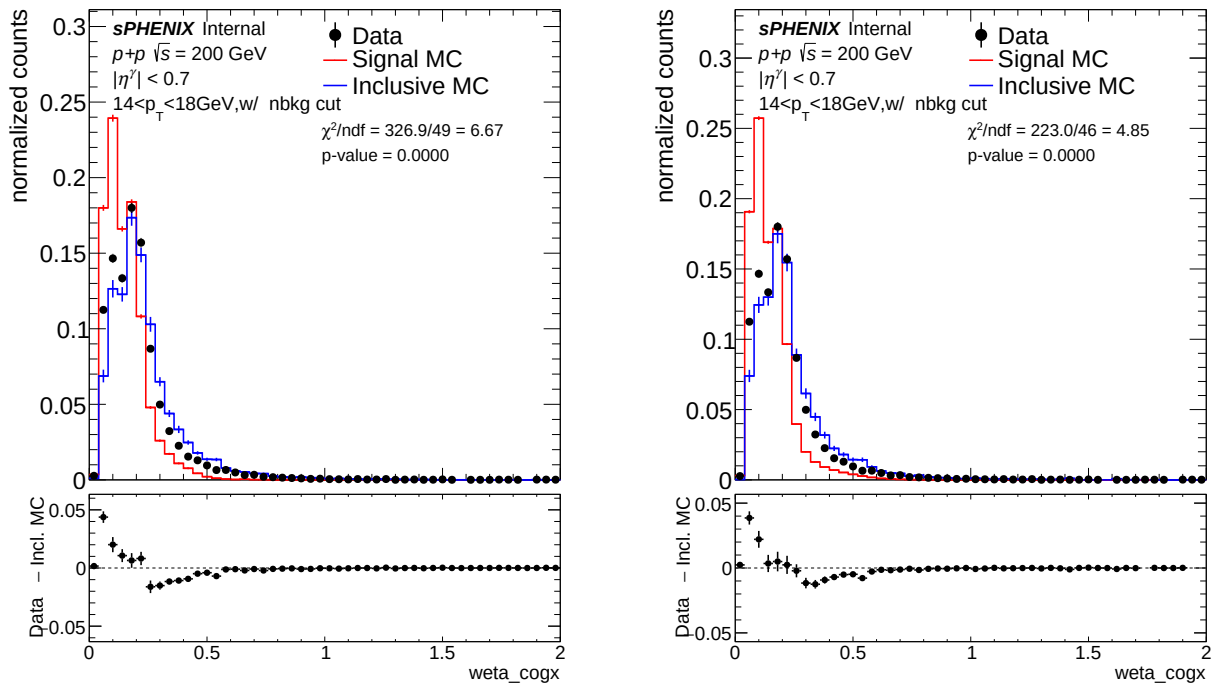
### 3.1 $w_\eta^{\mathrm{CoG}}$ (Eta Cluster Width)




(a) Nominal MC

(b) Mixed MC (7.9% DI)

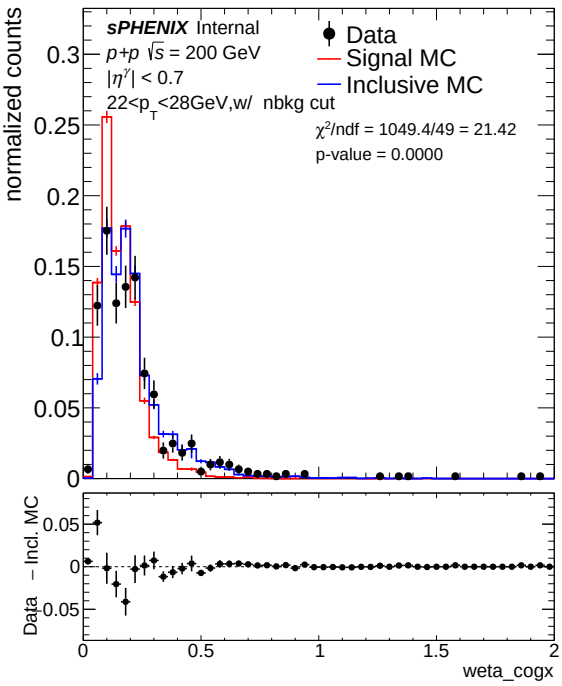
Figure 8: $w_\eta^{\mathrm{CoG}}$, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 9.72$; mixed $\chi^2/\mathrm{ndf} = 42.31$ (4.4× **degradation**).



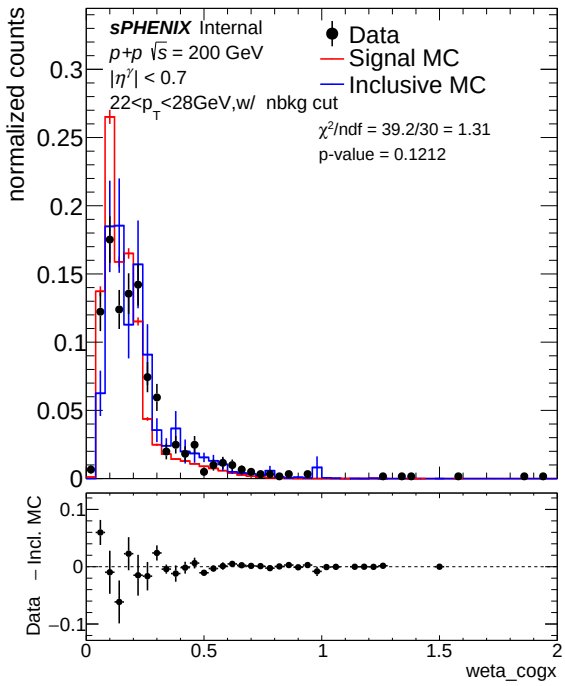

(a) Nominal MC

(b) Mixed MC (7.9% DI)

Figure 9: $w_\eta^{\mathrm{CoG}}$, $14 < E_T < 18\,\mathrm{GeV}$ (pt1), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 6.73$; mixed $\chi^2/\mathrm{ndf} = 4.86$ (1.4× improvement).



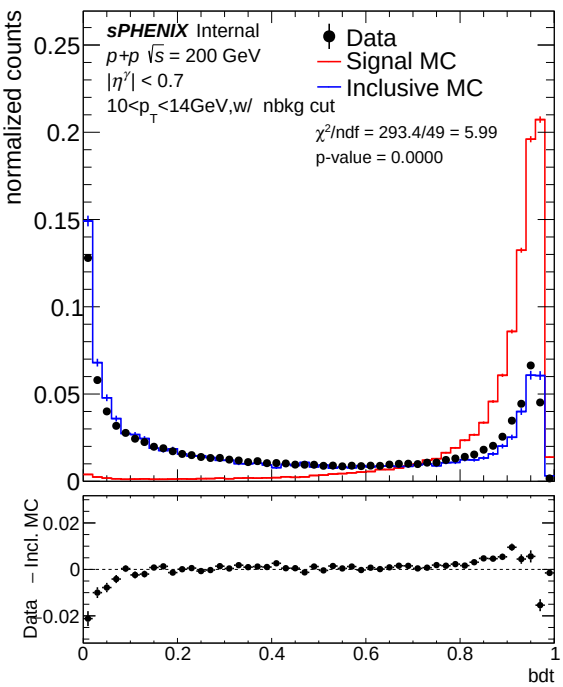

(a) Nominal MC



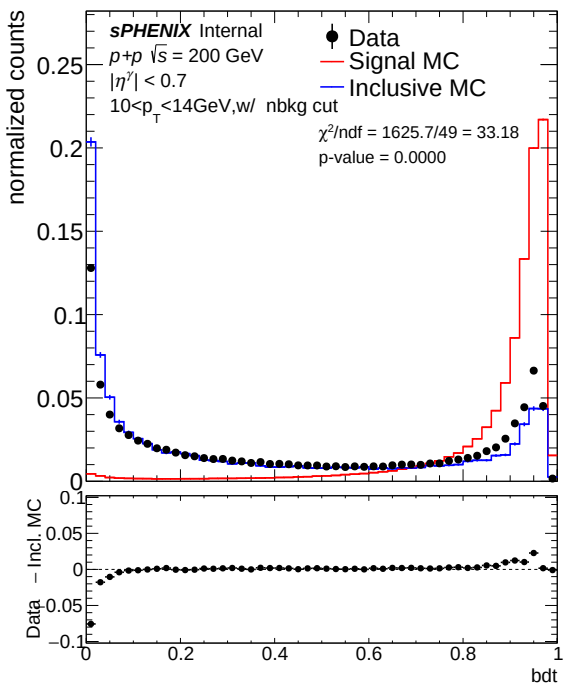

(b) Mixed MC (7.9% DI)

Figure 10: $w_\eta^{\mathrm{CoG}}$, $22 < E_T < 28\,\mathrm{GeV}$ (pt3), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 21.10$; mixed $\chi^2/\mathrm{ndf} = 1.31$ ($p = 0.12$; 16.1× improvement).

## 3.2 BDT Score

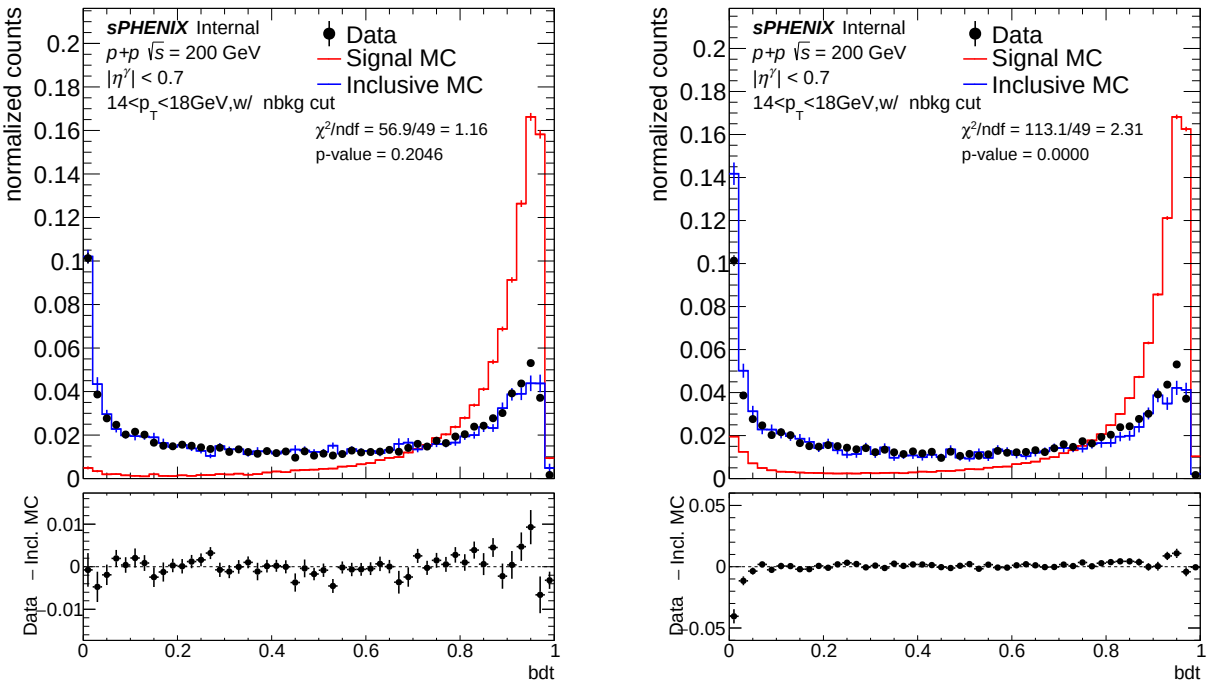



(a) Nominal MC




(b) Mixed MC (7.9% DI)

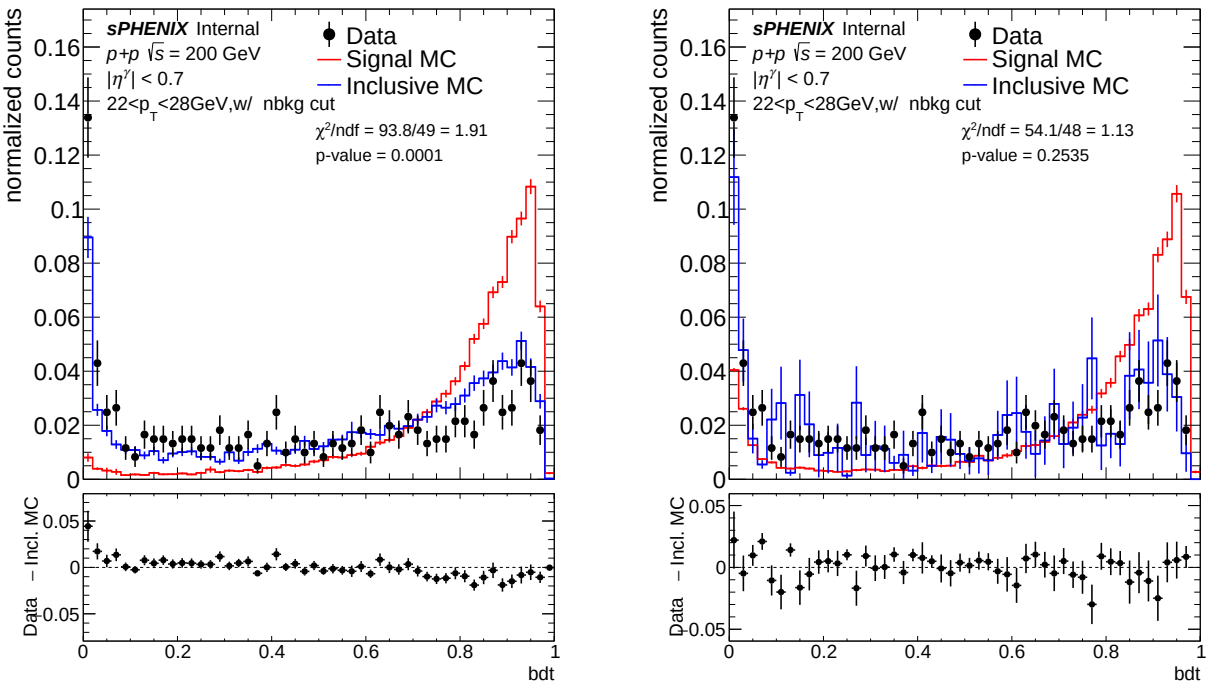
Figure 11: BDT score, $10 < E_T < 14\,\mathrm{GeV}$ (pt0), 1.5 mrad. Nominal $\chi^2/\mathrm{ndf} = 5.98$; mixed $\chi^2/\mathrm{ndf} = 33.08$ (5.5× **degradation**).




(a) Nominal MC

(b) Mixed MC (7.9% DI)

Figure 12: BDT score, $14 < E_T < 18\,\text{GeV}$ (pt1), 1.5 mrad. Nominal $\chi^2/\text{ndf} = 1.16$ ($p = 0.21$); mixed $\chi^2/\text{ndf} = 2.31$ (2.0× **degradation**; nominal already excellent).

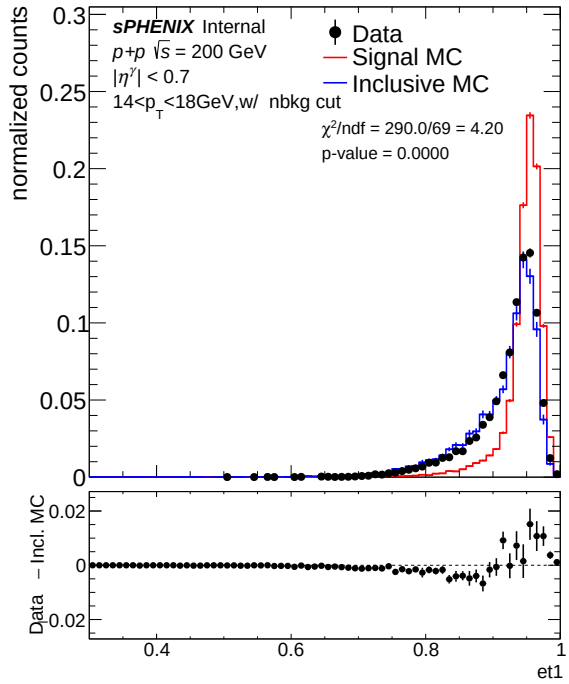



(a) Nominal MC

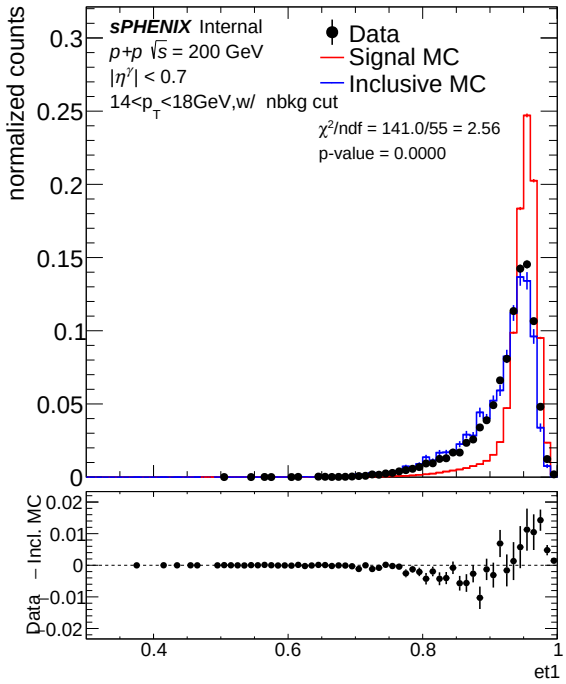
(b) Mixed MC (7.9% DI)

Figure 13: BDT score, $22 < E_T < 28\,\text{GeV}$ (pt3), 1.5 mrad. Nominal $\chi^2/\text{ndf} = 1.89$; mixed $\chi^2/\text{ndf} = 1.13$ ($p = 0.25$; 1.7× improvement).

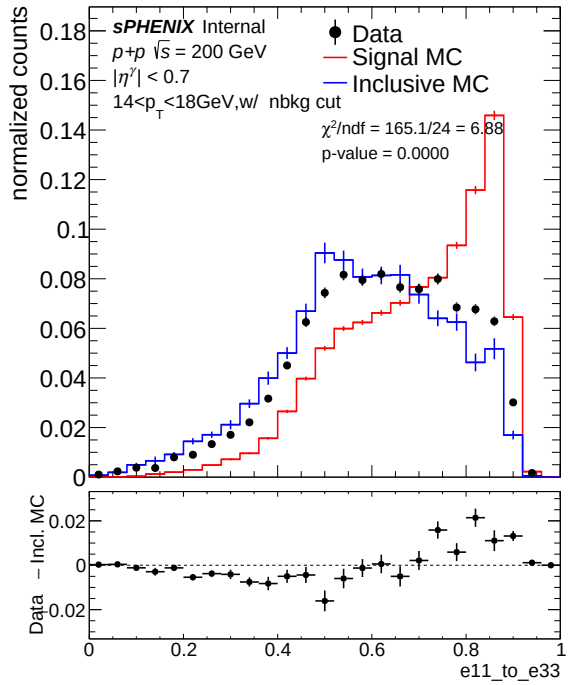
### 3.3 $f_1$ and $E_{1\times1}/E_{3\times3}$



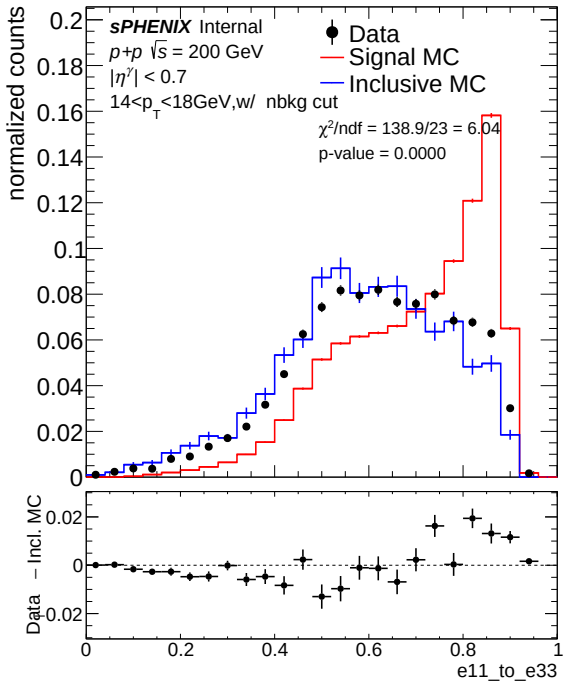

(a) $f_1$, Nominal MC




(b) $f_1$, Mixed MC (7.9% DI)




(c) $E_{1\times1}/E_{3\times3}$, Nominal MC

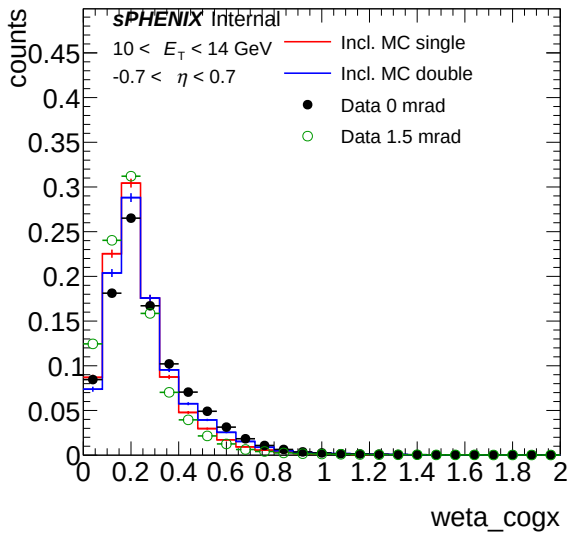



(d) $E_{1\times1}/E_{3\times3}$, Mixed MC (7.9% DI)

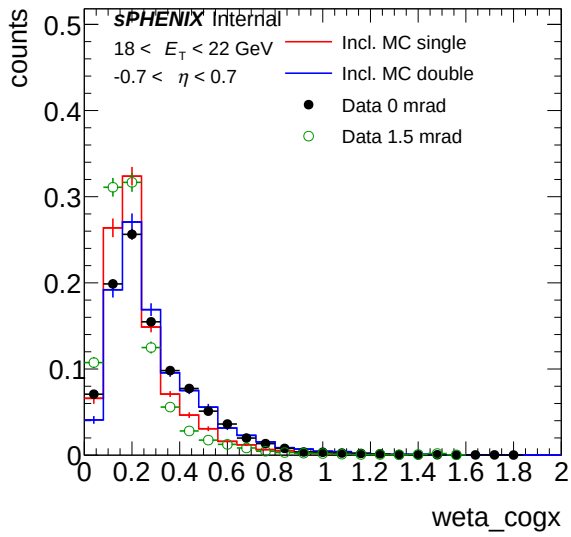
Figure 14: $f_1$ (top) and $E_{1\times1}/E_{3\times3}$ (bottom) at $14 < E_T < 18\,\text{GeV}$ (pt1), 1.5 mrad. $f_1$: $\chi^2/\text{ndf} = 4.17 \rightarrow 2.55$ (1.6×). $E_{1\times1}/E_{3\times3}$: $\chi^2/\text{ndf} = 6.88 \rightarrow 6.04$ (1.1×).

# 4 Single vs Double Interaction Overlay

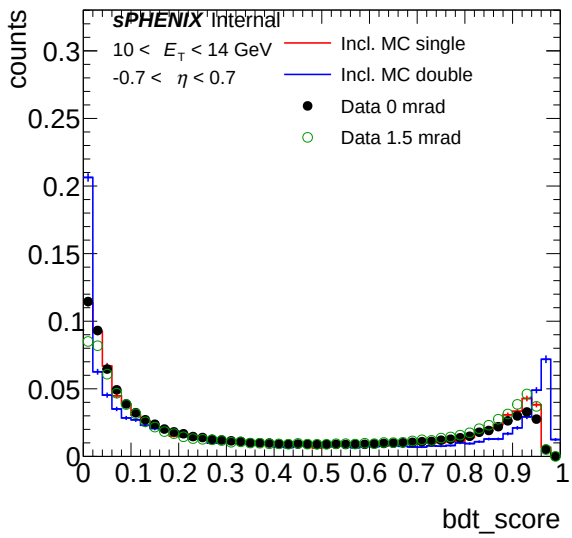
Figure 15 overlays single-interaction MC (red) and double-interaction MC (blue) with data from both crossing-angle periods: 0 mrad (filled black circles) and 1.5 mrad (open green circles), all area-normalized. Data at 0 mrad shows larger deviation from single MC, consistent with the higher pileup fraction, while data at 1.5 mrad tracks closer to single MC.



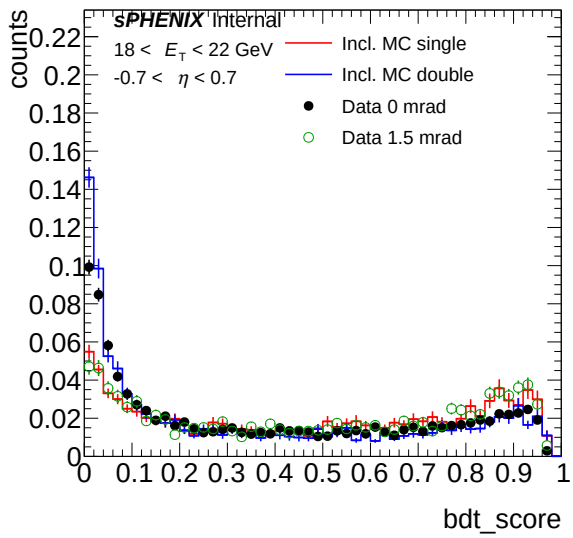

(a) $w_\eta^{\mathrm{CoG}}$, 10–14 GeV



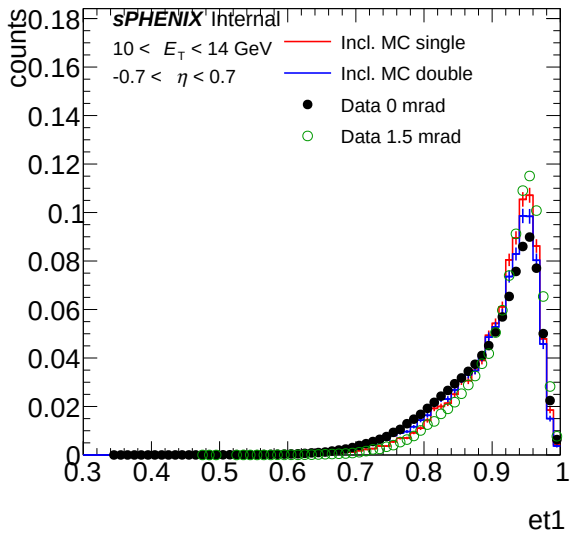

(b) $w_\eta^{\mathrm{CoG}}$, 18–22 GeV



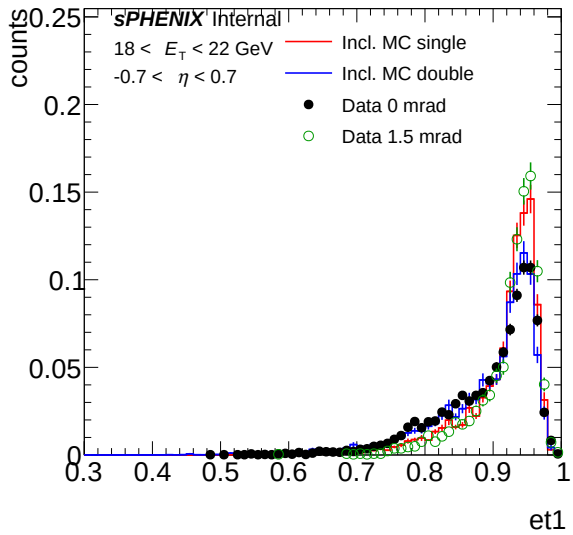

(c) BDT score, 10–14 GeV



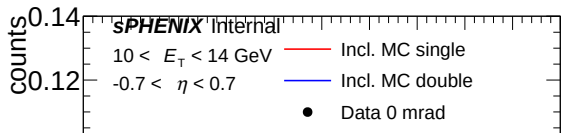

(d) BDT score, 18–22 GeV



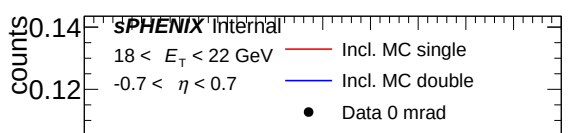

(e) $f_1$, 10–14 GeV




(f) $f_1$, 18–22 GeV





# 5 $\chi^2$/ndf Summary

Table 1 compiles the Data–Inclusive MC $\chi^2$/ndf and $p$-values for all variables, $E_T$ bins, and crossing angles. Entries where the mixed MC *degrades* agreement relative to nominal are highlighted in red. Improvement factors are computed as Nominal/Mixed when the mixed MC improves, and Mixed/Nominal (prefixed with a minus sign) when it degrades.

Table 1: Data–Inclusive MC $\chi^2$/ndf and $p$-values at cut1 for nominal (single-only) and mixed (double-blended) MC. Lower $\chi^2$/ndf is better; $p$-values $\ll 1$ indicate significant data–MC disagreement.

| | | **0 mrad (22.4% DI)** | | | | **1.5 mrad (7.9% DI)** | | | |
|---|---|---|---|---|---|---|---|---|---|
| Variable | $E_T$ [GeV] | Nom. | $p$ | Mixed | $p$ | Nom. | $p$ | Mixed | $p$ |
| $w_\eta^{\mathrm{CoG}}$ | 10–14 (pt0) | 47.40 | <0.001 | 28.96 | <0.001 | 9.72 | <0.001 | 42.31 | <0.001 |
| $w_\eta^{\mathrm{CoG}}$ | 14–18 (pt1) | 17.64 | <0.001 | 2.99 | <0.001 | 6.73 | <0.001 | 4.86 | <0.001 |
| $w_\eta^{\mathrm{CoG}}$ | 22–28 (pt3) | 15.58 | <0.001 | 0.75 | 0.840 | 21.10 | <0.001 | 1.31 | 0.118 |
| BDT score | 10–14 (pt0) | 6.85 | <0.001 | 20.30 | <0.001 | 5.98 | <0.001 | 33.08 | <0.001 |
| BDT score | 14–18 (pt1) | 12.20 | <0.001 | 1.63 | 0.004 | 1.16 | 0.207 | 2.31 | <0.001 |
| BDT score | 22–28 (pt3) | 9.62 | <0.001 | 1.75 | 0.001 | 1.89 | <0.001 | 1.13 | 0.249 |
| $f_1$ | 14–18 (pt1) | 9.33 | <0.001 | 1.92 | <0.001 | 4.17 | <0.001 | 2.55 | <0.001 |
| $E_{1\times1}/E_{3\times3}$ | 14–18 (pt1) | 4.50 | <0.001 | 4.14 | <0.001 | 6.88 | <0.001 | 6.04 | <0.001 |

# 6 Discussion

## 6.1 Universal pattern: $E_T$-dependent crossover

A consistent pattern emerges across *both* crossing angles: the double-interaction blending **degrades** data–MC agreement for the BDT score at low $E_T$ (pt0, 10–14 GeV) while **improving** it at higher $E_T$. This is now confirmed for both 0 mrad (3.0× BDT degradation at pt0) and 1.5 mrad (5.5× BDT degradation at pt0). For $w_\eta^{\mathrm{CoG}}$, the crossover occurs between pt0 and pt1: at 0 mrad, the pt0 $w_\eta^{\mathrm{CoG}}$ still improves (1.6×), but at 1.5 mrad it degrades (4.4×), indicating that the crossover pileup fraction for $w_\eta^{\mathrm{CoG}}$ lies between 7% and 19% at low $E_T$.

## 6.2 Crossing-angle comparison

The improvement factors scale with the pileup fraction: the 22.4% admixture at 0 mrad yields larger improvements (up to 20.8× for $w_\eta^{\mathrm{CoG}}$ at pt3) than the 7.9% admixture at 1.5 mrad (up to 16.1×). Conversely, the 1.5 mrad nominals are already much better than their 0 mrad counterparts (e.g., BDT pt1: 1.16 vs. 12.20), confirming that lower pileup naturally reduces the data–MC discrepancy.

## 6.3 Low-$E_T$ degradation mechanism

The low-$E_T$ degradation affects the BDT score most severely because the BDT amplifies subtle shower shape distortions through its nonlinear decision function. Contributing factors include:

1. **Shared MC samples**: the same `photon10_double` and `jet12_double` samples are used for both crossing angles; beam conditions in the MC may better match one period than the other.

2. **Low-$E_T$ sensitivity**: clusters near the 8–14 GeV threshold are most sensitive to vertex shifts and energy redistribution.

3. **Small-fraction noise**: at 7.9% mixing, any mismatch between the double MC and real pileup gets amplified relative to the small physical correction.

# 7 Conclusions and Recommendations

1. **0 mrad period**: the 22.4% double-interaction blending substantially improves data–MC agreement for pileup-sensitive variables ($w_\eta^{\mathrm{CoG}}$, BDT, $f_1$) at $E_T \geq 14\,\mathrm{GeV}$, with improvement factors up to 20.8×. At pt0 (10–14 GeV), the BDT score degrades by 3.0× while $w_\eta^{\mathrm{CoG}}$ still improves modestly.

2. **1.5 mrad period**: the 7.9% blending helps at high $E_T$ ($\geq 18\,\mathrm{GeV}$) but degrades both $w_\eta^{\mathrm{CoG}}$ and BDT at low $E_T$. The nominal agreement is already acceptable ($\chi^2/\mathrm{ndf} \lesssim 10$ for most variables).

3. **Recommended baseline for cross-section measurement**:

   - **0 mrad,** $E_T \geq 14\,\mathbf{GeV}$: use mixed MC (22.4% DI)
   - **0 mrad,** $E_T < 14\,\mathbf{GeV}$: use mixed MC for $w_\eta^{\mathrm{CoG}}$; use nominal MC for BDT-sensitive quantities; assign nominal–mixed difference as systematic
   - **1.5 mrad,** $E_T \geq 18\,\mathbf{GeV}$: use mixed MC (7.9% DI)
   - **1.5 mrad,** $E_T < 18\,\mathbf{GeV}$: use nominal MC; assign nominal–mixed difference as systematic

4. **Alternative**: use nominal MC for the full 1.5 mrad period and apply the double-interaction correction to 0 mrad data only.